

the trade (Almgren and Chriss, 2000). The AIM adaptation tactic becomes more aggressive in times of favorable price momentum and less aggressive in times of adverse price momentum.

If investors selected the price improvement best execution strategy in step 2 it is essential that we use the exact same cost in the AIM adaptation strategy, otherwise the resulting strategy will have a lower probability of executing more favorably than initially intended and will not be consistent with the trading goal.

Mathematically, the AIM tactic is found by maximizing the following equation:

$$\begin{aligned} \text{Min } L &= \frac{E_0(\text{Cost}) - E_t(\text{Cost})}{E_t(\text{TR})} \\ \text{s.t. } LB^* &\leq \alpha_t \leq UB^* \end{aligned}$$

Here, $E_0(\text{Cost})$ is either the original expected cost from the BES in step 2 or the cost used to generate the price improvement strategy in step 2. And $E_t(\text{Cost})$ and $E_t(\text{TR})$ are the expected projected cost and timing risk for the order at time t . Also, the expected cost term needs to include the alpha cost component in times when traders have a short-term alpha expectation.

For the trading rate strategy (without an alpha term) the AIM adaptation tactic objective function is:

$$\text{Min } L = \frac{C^* - (K_t + (1 - \theta)(b_1 \cdot I' \cdot \alpha_t))}{\sigma \cdot \sqrt{(1 - \theta) \cdot \frac{X}{ADV} \cdot \frac{1}{\alpha_t} \cdot P_0}} \quad (8.37)$$

Where C^* is either the original expected cost from the BES or the cost used to generate the price improvement best execution strategy and K_t is the unavoidable cost at time t based on realized cost and market movement.

Solving for α_t we get,

$$\alpha_t = \frac{C^* - K_t}{3 \cdot (1 - \theta)(b_1 \cdot I')} \quad (8.38)$$

This trading rate then needs to ensure that it satisfies our boundary conditions (i.e., user specified maximum and/or minimum rates, or the minimum rate required to ensure completion of the order). That is written mathematically as:

$$\alpha_t^* = \min(\max(LB, \alpha_t), UB)$$